

Inequalities

These are ordered in (roughly) ascending difficulty.

Problem 1

Show that $\mathbb{E}[\ln(X)] \leq \ln(\mathbb{E}[X])$ for any discrete non-negative random variable X .

Problem 2

Prove that probability of getting doubles when rolling a pair of (identical, but possibly unfair) n -sided die is at least $1/n$.

Problem 3

Let a, b, c be positive real numbers such that $abc = 1$. Prove that:

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}$$

Problem 4

Let $n \geq 3$ be an integer and $a_2, a_3, \dots, a_n$ positive real numbers such that $a_2 a_3 \dots a_n = 1$. Prove that

$$(a_2 + 1)^2 (a_3 + 1)^3 \dots (a_n + 1)^n > n^n$$

Problem 5

Let $a, b, c, d > 0$ be positive real numbers such that $abcd = 1$ and $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} < a + b + c + d$. Prove that $\frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d} > a + b + c + d$.